

Amphenol®



SurLok Plus™ Series

www.amphenol-industrial.com

What is SurLok Plus™?



The SurLok Plus™ compression lug is a field installable, highly reliable alternative to common compression lugs

Using industry standard crimp, screw, and busbar termination options, thus eliminating the need to purchase special torque tools.

SurLok Plus™ is an environmentally sealed version of our original SurLok® but available in smaller sizes, and features a quick lock and press-to-release design. Incorporating the latest R4 RADSOK® Technology. SurLok Plus™ is a compact, quick mating, and robust product line.

The RADSOK® high amperage contact technology utilizes high tensile strength properties of a stamped and formed, high conductivity alloy grid to produce low insertion forces while maintaining a large conductive surface area. The R4 version of the RADSOK® represents the culmination of three years of research and development in laser welding copper based alloys.

Markets

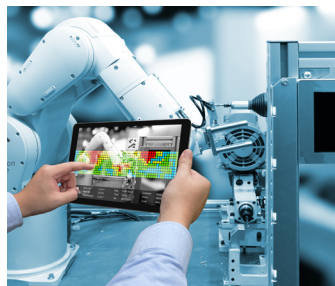
The markets and applications for the SurLok Plus™ Series include but are not limited to the following:



Energy Storage



EV/HEV*



Process Control/Automation



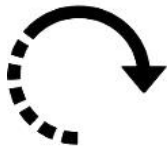
Heavy Equipment

*Note: 1500V SurLok Plus™ option only

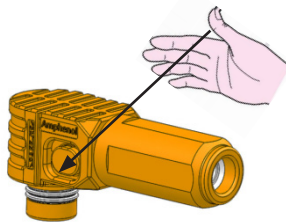
Features and Benefits

Keyway	Multiple keyways
Lock	Quick lock and press-to-release design
Plug	360° rotating plug optional
Receptacle	Various termination options (Threaded, Crimp, Busbar)
Voltage	1000V / 1500V optional
Protection	Touch Proof
Design	Compact robust design

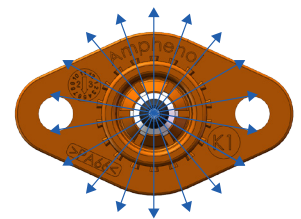
- The connector's design incorporates an integral latching system that ensures a definitive electrical and mechanical connection.
- Connector housings are manufactured with a thermoplastic material that is not only durable, but also with excellent dielectric, mechanical properties and meet environmentally RoHS compliant.
- Contacts are derived from quality copper alloy to ensure an electrically-reliable connection.
- UL1977/4128 and TUV 2PFG2740 certified



Rotation

Quick lock and
press-to-release design

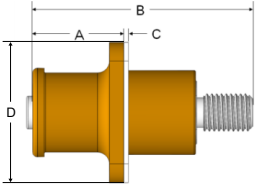
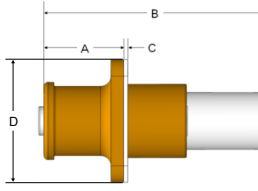
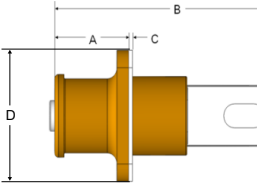
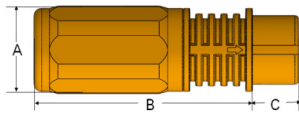
1000/1500 VDC

Multi-Directional Mating
Positions

Technical Information

Description	RADSOX® Size						
	3.6mm	5.7mm	8.0mm	8.0mm EMI	10.3mm	12.0mm	14mm EMI
Current Rating	50A~70A	100A~120A	120A~250A	150A~250A	250A~300A	300A~360A	300A~500A
Operating Temperature	-40°C~125°C (1500V DC)	-40°C~105°C (1000V DC)	-40°C~105°C (1000V DC)	-40°C~125°C (1000V DC)	-40°C~105°C (1000V DC)	-40°C~125°C (1500V DC)	-40°C~140°C (1500V DC)
		-40°C~125°C (1500V DC)	-40°C~125°C (1500V DC)		-40°C~125°C (1500V DC)		
IP Sealing(Mated)	IP67/ IP6K9K (1500V DC)	IP67 (1000V DC)	IP67 (1000V DC)	IP67	IP67 (1000V DC)	IP67/IP6K9K	IP67/IP6K9K
		IP67/IP6K9K (1500V DC)	IP67/IP6K9K (1500V DC)		IP67/IP6K9K (1500V DC)		
Flammability	UL 94 V-0	UL 94 V-0	UL 94 V-0	UL 94 V-0	UL 94 V-0	UL 94 V-0	UL 94 V-0
Locking	Spring Locking	Spring Locking	Spring Locking	Secondary Lock	Spring Locking	Spring Locking	CPA
RoHS Compliant	Yes	Yes	Yes	Yes	Yes	Yes	Yes
EMI Shielding	No	No	No	Yes	No	No	Optional
HVIL Function	No	No	No	Yes	No	No	Optional
Durability	≥100 cycles	≥100 cycles	≥100 cycles	≥100 cycles	≥100 cycles	≥100 cycles	≥100 cycles
IPXXB	Yes	Yes	Yes	Yes	Yes	Yes	Yes

1000V DC Product Dimensions - Receptacle

Item	Screw Termination				Crimp Termination				Busbar Termination				Inline Crimp Termination		
Size															
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C
5.7mm	16.5	36.7	1.0	27	16.5	40.7	1.0	27	16.5	46.7	1.0	27	20	53.7	8.8
8.0mm	19.0	46.2	1.0	30	19.0	52.7	1.0	30	19.0	61.2	1.0	30	24.3	56.7	12
10.3mm	25.8	55.5	0.3	35	25.8	50mm²: 65.8	0.3	35	25.8	68.8	0.3	35	/	/	/
						70mm²: 68.8									
						85mm²: 71.8									
						95mm²: 71.8									

Note: All dimensions for reference only

Ordering Information - 1000V DC Receptacle

Product Series	Type		RADSOK® Size		Termination Style		Panel Mount	Sealing		Color		Keyway****						
SLP	R	Receptacle W/O Keyway	A	5.7mm 120A	B	Busbar	P	W/ Flange	S	IP67	O	Orange	0	NO KEYWAY				
					T	Threaded												
					16C	16mm² Crimp (100A max.)												
					25C	25mm² Crimp (120A max.)												
			B	8.0mm 200A	B	Busbar					N	W/O Flange**	N	Non-Sealed	B	Black	1	180°
					T	Threaded												
					25C	25mm² Crimp (120A max.)												
					35C	35mm² Crimp (150A max.)												
	IR	Receptacle W/ Keyway	C	10.3mm 300A	50C	50mm² Crimp (200A max.)	N	W/O Flange**	N	Non-Sealed	B	Black	2	60°				
					B	Busbar												
					T	Threaded												
					50C	50mm² Crimp (200A max.)												
			CR	*Inline Receptacle	A	5.7mm 120A	70C	70mm² Crimp (250A max.)	B	W/ Back shells***	N	Non-Sealed	R	Red	3	120°		
							85C	85mm² & 95mm² (300A max.)										
							16	16mm² (100A max.)										
							25	25mm² (120A max.)										
B	8.0mm 200A	25			25mm² (120A max.)	B	W/ Back shells***	N	Non-Sealed	R	Red	4	150°					
		35			35mm² (150A max.)													
		50			50mm² (200A max.)													

*Inline receptacle, 90° and 120° keyways only ** Not available for 10.3mm ***Only available for inline type **** 10.3mm requires Keyway selection

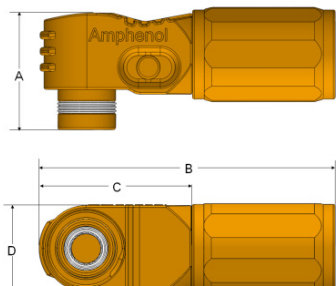
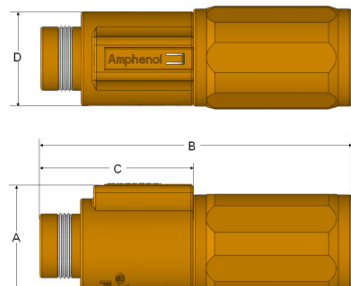
Example:

1	2	3	4	5	6	7	8
Product Series	Type	RADSOX® Size	Cable Size	Back shells	Sealing	Color	Keyway
SLP	IR	A	25	B	S	O	3
SLP	CR	B	50	B	N	B	0

SLPIRA25BSO3: 5.7mm SurLok Plus™ Receptacle without shielding and no HVIL, orange, 25mm² cable crimp, waterproof: sealed, keyway: 120°

SLPCRB50BNB0: 8.0mm SurLok Plus™ Inline receptacle without shielding and no HVIL, black, 50mm² cable crimp, waterproof: non-sealed, keyway: 90°

1000V DC Product Dimensions - Plug

Item	Right Angle Plug				Straight Plug			
Size								
	A	B	C	D	A	B	C	D
5.7mm	22.6	55.1	30	18	22.5	63	30.5	20
8.0mm	29	74.5	38.4	21.6	24.45	72.4	34.8	21.4
10.3mm	41.2	83.5	42.5	28	/	/	/	/

Note: All dimensions for reference only

Ordering Information - 1000V DC Plug

Product Series	Type		RADSOK® Size		Cable Size		Back Shells		Sealing		Color		Keyway****					
SLP	P	Right Angle 90°	A	5.7mm 120A	16	16mm² (100A max.)	B	W/ Back Shells	S	Sealed	O	Orange		NO KEYWAY				
					25	25mm² (120A max.)								0	90°			
			B	8.0mm 200A	25	25mm² (120A max.)							B			Black	1	180°
					35	35mm² (150A max.)												
					50	50mm² (200A max.)								3	120°			
			C	10.3mm 300A	50	50mm² (200A max.)					R	Red	4			150°		
					70	70mm² (250A max.)												
					85	85mm² & 95mm² (300A max.)												
	IP	Straight Type	A	5.7mm 120A	16	16mm² (100A max.)	N	W/O Back Shells	N	Non-Sealed								
					25	25mm² (120A max.)												
			B	8.0mm 200A	25	25mm² (120A max.)												
					35	35mm² (150A max.)												
					50	50mm² (200A max.)												

*Inline receptacle, 90° and 120° keyways only ** Not available for 10.3mm ***Only available for inline type **** 10.3mm requires Keyway selection

Example:

1	2	3	4	5	6	7	8
Product Series	Type	RADSOK® Size	Cable Size	Back shells	Sealing	Color	Keyway
SLP	P	A	25	B	S	O	
SLP	IP	A	25	B	S	O	3

SLPPA25BSO: 5.7mm SurLok Plus™ Right Angle Plug without shielding and no HVIL, orange, 25mm² cable, waterproof: sealed, keyway: NO KEYWAY

SLPIPA25BSO3: 5.7mm SurLok Plus™ Straight Plug without shielding and no HVIL, orange, 25mm² cable crimp, waterproof: sealed, keyway: 120°